



Project Specification
Implementation of
HYDRA

at



- Interface Concept -

Version 1

Date: 21.05.2019

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Declaration of Approval

Specification for the
Interface ERP-HYDRA at [fram as Kunststofftechnik GmbH](#)
Version: 1
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1 General

This document explains and defines the necessary interfaces between ERP WinLine and MES HYDRA.

This document refers occasionally to other documents (product documentation or EXCEL file) for further information and to prevent duplicate descriptions.

In a further project step this document may be included in the general specification.

1.1 Project language (language for customizations)

Project language is English. For simplicity interface documentations are provided in German and English.

1.2 Glossary and abbreviations

Term/abbreviation	Description
AIP	Acquisition and Information Panel (AIP) - the shop floor terminal of MES
BDE	Shop Floor Data Collection
CONFIG	"Standard" HYDRA configuration made by the customer after participating in required HYDRA training courses. If required, an MPDV consultant supports the customer.
CUSTOM	Customization developed and delivered by MPDV
DMC	Dynamic Manufacturing Control
DNC	Decentral NC management
EIS	Enterprise Integration Services
EMG	Energy Management
FAT	Factory Acceptance Test
FEP	In-Production Inspection
FMEA	Failure Mode and Effects Analysis
HLS	Shop Floor Scheduling for detailed planning
INFO	Information on an issue or chapter in this document.
LLE	Incentive Pay
MDE	Machine Data Collection
MDS	MES Development Suite
MIS	MES Implementation Specification
MOC	MES Operation Center (MOC) - central GUI of the MES HYDRA at workstation PCs
MRS	MES Requirement Specification
MPL	Material and Production Logistics
n/a	Not applicable (not relevant)
OP	Operation/process of an order (Usually, this is the combination of order + operation in HYDRA)
PCC	Process Communication Controller PCC is software consisting of different components and connecting machines
PDM	Production Data Manager (Data communication using "HYDRA dialog data")
PDV	Process Data Collection

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PEP	Personnel Scheduling
PMV	Gage Management
PZE	Attendance and Personnel Time Management
PZW	Personnel Time Management
QMS	Quality Management System
REK	Complaints Management
SAT	Site Acceptance Test
SIF	Service interface (Data communication using webservice)
SIS	System Integration Services
SMA	Smart Mobile Applications
STD	Standard functionality and, if necessary, required MES HYDRA licenses
TRT	Tracking & Tracing
WEP	Incoming Goods Inspection
WRM	Tool and Resource Management
ZKS	Access Control

1.3 Further applicable documents

Refer here to other documents (e.g. the relevant project order or minutes, e-mails, interface specifications, etc.). If possible, you should insert links to the documents.

File name	Version	Description
FRAM_interface_datamapping_17052019.Excelx	1	EXCEL file which explains fields in interface segments.

1.4 Stakeholders

Person (company) & contact details, if necessary	Description
Kai Frank, Sascha Berger, Fabian Sprau	framass Kunststofftechnik
Roman Gaidies	Mesonic software GmbH (supplier of ERP software WinLine)

1.5 Overview of OPEN issues

Es wurden keine Einträge für das Inhaltsverzeichnis gefunden.

1.6 Overview of CUSTOM issues

Es wurden keine Einträge für das Inhaltsverzeichnis gefunden.

1.7 Overview of CONFIG issues

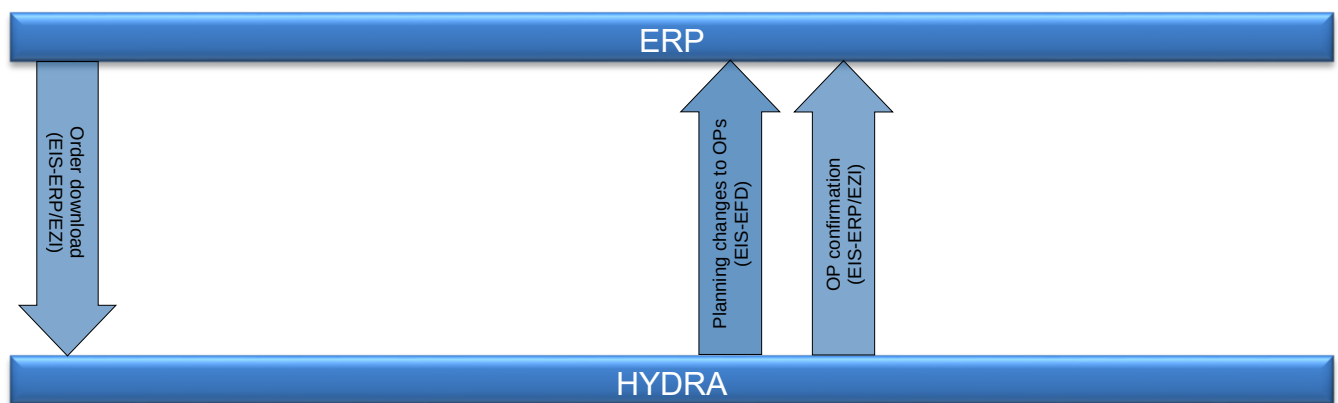
CONFIG FRAM-003-002-001: BAPINOUPDATE	16
CONFIG FRAM-004-002-003: Operation planned	18
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2 System context

2.1 Overview of interface (schematic)

Following interfaces will be used to connect ERP WinLine to MES HYDRA. All interfaces are based on standard HYDRA definition. Instead of the file based implementation the database interface will be used, file generation will be turned off during interface setup.

Used interfaces are EIS-ERP + EIS-EZI for order/operation download and upload of timetickets. Interface EIS-EFD will be used to upload scheduling information from HYDRA to ERP. To connect both system the database interface EIS-DBI is used.



2.2 Order/Routing structure in ERP

A typical order structure at FRAM consists of one order header with one or multiple operations. Each operation will provide its own components. In the first steps no MPL/TRT is used and for this reason the components are mainly for information in HYDRA.

In addition parallel order sequences are used. The necessary sequencing information is provided by ERP via interface.

2.3 Parallel sequences

To process orders and operations as required the functionality of parallel sequences will be used. To process parallel sequences in HYDRA the license BDE-APF is required. This license is at the moment not included in the project scope.

LIC FRAM-001: necessary licenses

BDE-APF (Processing of alternative/parallel sequences)

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TODO FRAM-001-001: Start the order with parallel sequences?

According to documentation BDE-APF only parallel sequences after one “normal” operation are mentioned. Is it possible to start an order with a parallel sequence?

Responsible: MPDV, Weiß / date: 21.05.2019

Weiß, 20.05.2019: If the data is provided correct and in full, it's possible to start an order with parallel sequences



TODO FRAM-001-002: Multiple sequences per order

Is it possible to use multiple sequences per order?

Responsible: MPDV, Weiß / date: 21.05.2019

Weiß, 20.05.2019: It's possible to have multiple sequences per order, see also documentation BDE-APF.

3 Interfaces

3.1 Interface FRAM-002: Database interface instead of file based interface

3.1.1 Function FRAM-002-001: Database interface EIS-DBI

Background information:

The background information serves to better understand the requirements of FRAM.
This information is not relevant for the implementation of the requirements.

FRAM wants to use the database interface to transfer data between ERP and MES. No files should be transferred between systems. To fulfill this requirement FRAM wants to implement the database interface EIS-DBI to access the interface database tables for inbound and outbound data transfer.

In following chapters the use of the database interface is taken as prerequisite and not described each time. Transfer of data and further information in regard to the database interface is handled in this chapter only.

STD FRAM-002-001-001: EIS-DBI

HYDRA standard interface EIS-DBI will be used to transfer data between ERP and MES. No file based interface transfer will take place.

LIC FRAM-002-001-002: necessary licenses

EIS-DBI (Interface based on databases)

Functions in MES HYDRA:

The database interface EIS-DBI provides the customer access to the data staging tables for inbound and outbound data transfer. It's possible to insert and read data from these DB tables. For correct processing of the data processing information need to be created by the external system (in this case ERP) which would normally be created by HYDRA. These information are mainly control records in addition to the required data records.

Access to the application database tables is not permitted by license EIS-DBI!

3.1.1.1 Activation

The DB interface will be activated based on the license EIS-DBI and the interface setup.

3.1.1.2 Transfer media

Data is transferred from ERP to MES into the staging DB tables.

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3.1.1.3 Data structure

The data structure of each involved interface is described in the corresponding chapter and linked documentation.

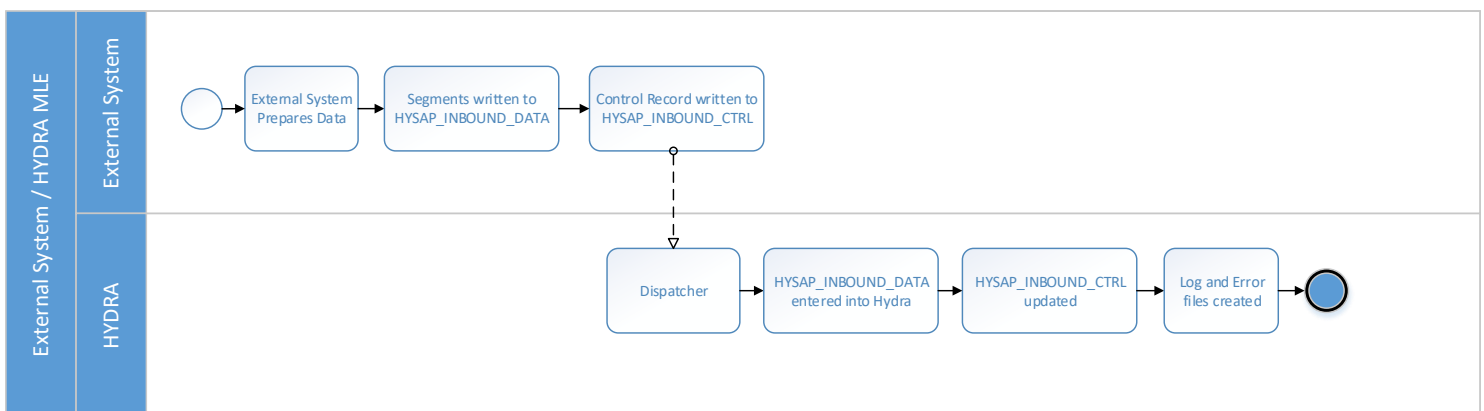
3.1.1.4 Archiving

- Interface data of MLE inbound transactions ("HYSAP-INBOUND") are archived after two days and deleted after seven days (configured in HYDRA distribution model).
- If required (especially during the test phase), you can change this interval for every message type.

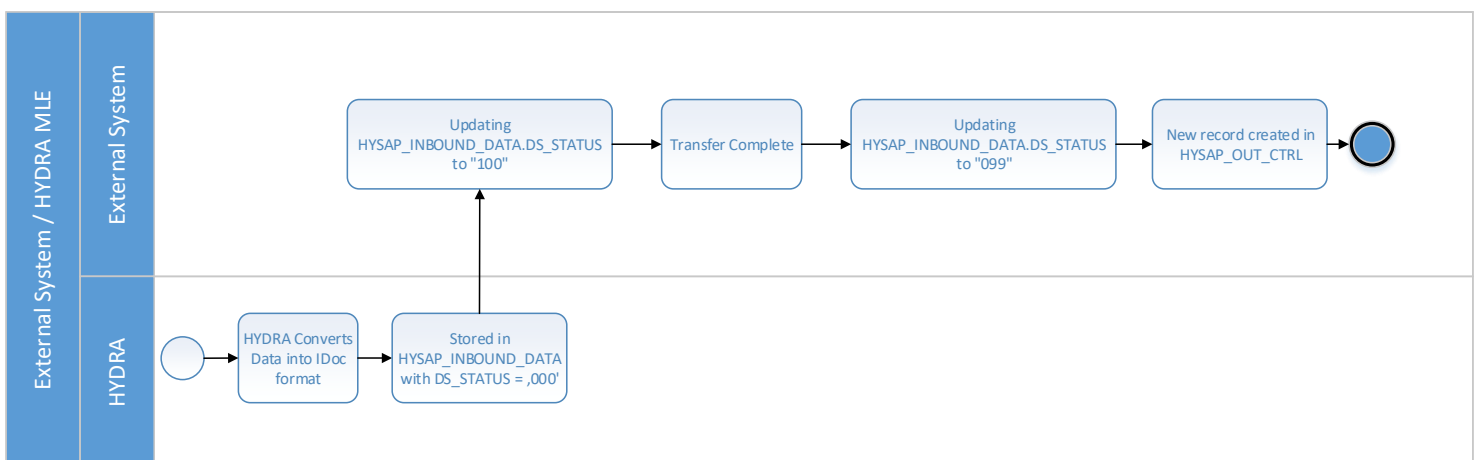
3.1.1.5 Mode of operation

Following the general workflow and responsibilities of the Systems are visualized.

For inbound transfer (ERP --> HYDRA):



For outbound transfer (HYDRA --> ERP)



Further information can be found in documentation "EIS-DBI_30.pdf" and in the EXCEL file "FRAM_interface_datamapping_17052019.xlsx" provided by Michael Weiß. In the documentation the transfer and DB table columns are described in detail.

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3.1.1.6 Expected result

Data records and control records are transferred correct from ERP to HYDRA and HYDRA to ERP via DB interface.

3.2 Interface FRAM-003: Interface from ERP to HYDRA

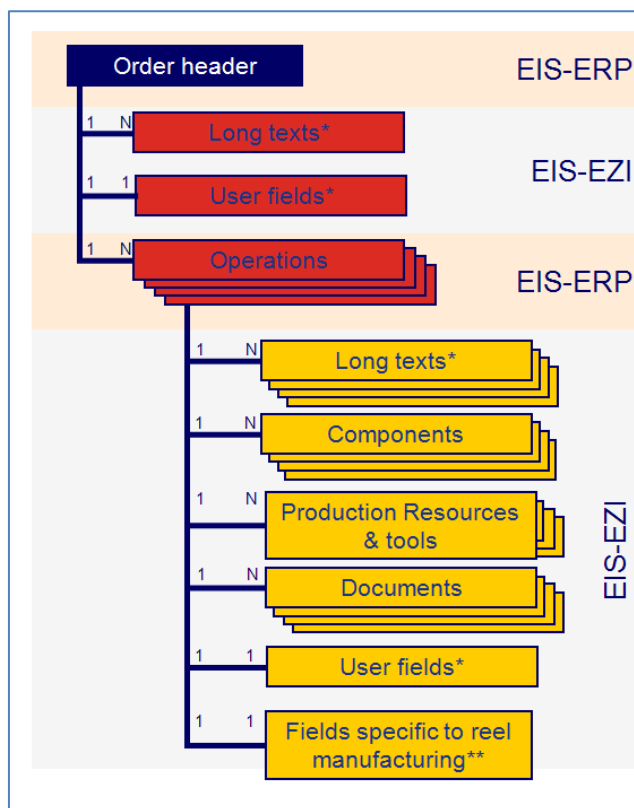
3.2.1 Function FRAM-003-001: Order and operation interface EIS-ERP + EIS-EZI

Background information:

The background information serves to better understand the requirements of FRAM.
This information is not relevant for the implementation of the requirements.

With the interface EIS-ERP and extension EIS-EZI information for orders and operations are transferred from ERP to MES. In following descriptions the name EIS-ERP is used for the combined interface licenses EIS-ERP and EIS-EZI.

Following image visualizes the hierarchy of transferrable data and the relation between them.



STD FRAM-003-001-001: EIS-ERP + EIS-EZI

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HYDRA standard interface EIS-ERP + EIS-EZI will be used.

Data will be transferred via DB interface EIS-DBI instead of standard file based interface.

Functions in MES HYDRA:

Necessary order/operation data will all be transferred via interface EIS-ERP.

3.2.1.1 Activation

The interface is activated during interface setup and based on the licenses EIS-ERP/EZI.

3.2.1.2 Transfer media

Data will be transferred via DB interface, see chapter 3.1.

3.2.1.3 Data structure

The general data structure of the interface is as follows. Note: not all segments are mandatory.

Message type/

file name: HY72PPS

Message functions/

file extensions: DAT

Segments:

- HY72_AU_HD_001_A (order header)
 - | HY72_AU_INFO_AI_001_A (long texts)
 - | L HY72_AU_USRFLD_001_A (user fields)
- HY72_AFOLG_001_A (Sequence)
- HY72_AG_HD_001_A (operation data – part 1)
 - | HY72_AG_HD_002_A (operation data – part 2)
 - | HY72_AG_KOMPL_001_A (component list)
 - | L HY72_AG_KOMPL_USRFLD_001_A (comp. user fields)
 - | HY72_AG_FHM_001_A (PRT / resources)
 - | HY72_AG_DOC_001_A (documents)
 - | HY72_AG_INFO_AI_001_A (long texts)
 - | HY72_AG_USRFLD_001_A (user fields)
 - | L HY72_AG_RF_001_A (MPL-RF-specific data)
- L HY72_FERTVAR_001_A (production variants)

(Taken from documentation EIS-ERP_81.pdf)

The single segments with data fields are explained in documentation EIS-ERP_81.pdf and the additional EXCEL file for the data mapping. The EXCEL file includes also example data, reference to ERP WinLine datafields and further explanations.

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3.2.1.4 Involved objects

Order Header (Segment: HY72_AU_HD_001_A)

Long texts for order header (Segment: HY72_AU_INFO_AI_001_A)

User fields for order header (Segment: HY72_AU_USRFLD_001_A)

Operation sequence (Segment: HY72_AFOLG_001_A)

Operation Data (Segment: HY72_AG_HD_001_A)

Componenten list for consumption materials (Segment: HY72_AG_KOMPL_001_A)

User fields for components (Segment: HY72_AG_KOMPL_USRFLD_001_A) (optional)

PRT / resource information (Segment: HY72_AG_FHM_001_A) (optional)

Links to documents on network share etc. (Segment: HY72_AG_DOC_001_A)

Long texts for operations (Segment: HY72_AG_INFO_AI_001_A) (optional)

User fields for operations (Segment: HY72_AG_USRFLD_001_A) (optional)

3.2.1.5 Mode of operation

Order/operation data will be provided by the ERP to MES. ERP will stage the data according to process for the DB interface EIS-DBI. Each segment results in one line of data in the database table.

Please note the description for create/modify and deletion of data via interface (chapter 6 "Order data ERP --> HYDRA" in documentation EIS-ERP_81.pdf).

INFO FRAM-003-001-002: Order sequences

For the correct processing of order sequences HYDRA needs the information for master sequence and parallel sequence via the interface. This information needs to be transferred for each order individually.

3.2.1.6 Expected result

Data for orders/operations is transferred to HYDRA and available for further processing.

3.2.2 Function FRAM-003-002: Protect fields during operation update via interface

Background information:

The background information serves to better understand the requirements of <CUSTOMER>.

This information is not relevant for the implementation of the requirements.

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If an operation is updated via interface and HYDRA HLS is used it's necessary to configure the functionality of "BAPINOUPDATE". This configuration prevents fields from updated via interface, this is necessary e.g. for the information on which workplace a specific operation is planned.

To use the BAPINOUPDATE functionality see documentation EIS-ERP, chapter 11. Following configuration should be used:

Example: protect the planned workplace

If the operation is planned on a workstation, you have to prevent the ERP interface from cancelling this planning and putting the operation back into the operation pool of workplaces. To do so, enter the below-mentioned data:

Field name	Value
Section	ANR
Key	ANR.MGRP@ANR.MNR@ANR.OPT:PLAN@
Value	ANR.ATYP=AG@ANR.OPT:PLAN=M@
Active	Yes

CONFIG FRAM-003-002-001: BAPINOUPDATE

Configure BAPINOUPDATE functionality according to documentation EIS-ERP to protect the planned workplace and further fields as needed.

3.3 Interface FRAM-004: Interface from HYDRA to ERP

3.3.1 Function FRAM-004-001: Order/operation bookings (timeticket) EIS-ERP

Background information:

The background information serves to better understand the requirements of FRAM

This information is not relevant for the implementation of the requirements.

Recorded quantities (yield, scrap with reason) and times will be uploaded via timetickets. HYDRA will provide this confirmations in the staging DB tables for outbound processing.

STD FRAM-004-001-001: EIS-ERP + EIS-EZI

HYDRA standard interface EIS-ERP + EIS-EZI will be used.

Data will be transferred via DB interface EIS-DBI instead of standard file based interface.

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3.3.1.1 Activation

Upload of confirmations is based on configuration in HYDRA at order types.

3.3.1.2 Transfer media

Data will be transferred via DB interface, see chapter 3.1.

3.3.1.3 Data structure

The general data structure of the interface is as follows.

Message type: HY72ADRCK_TT

IDOC type: HY72ADRCK_TT

The segment with data fields is explained in documentation EIS-ERP_81.pdf and the additional EXCEL file for the data mapping. The EXCEL file includes also example data, reference to ERP WinLine datafields and further explanations.

3.3.1.4 Involved objects

Recorded quantities and processing time are the base for the timetickets.

3.3.1.5 Properties

Each operation may result in several timetickets based on handling at the terminal. Each timeticket will be transferred, an aggregation of two or more timetickets into one timeticket will not take place.

3.3.1.6 Archiving

- Interface data of MLE outbound transactions ("HYSAP-OUTBOUND") are archived after two days and deleted after seven days (configured in HYDRA distribution model).
- If required (especially during the test phase), you can change this interval for every message type.

3.3.1.7 Mode of operation

HYDRA creates timetickets based on recorded order bookings which in turn are based on postings at the shopfloor terminal. Timetickets are provided in the outbound interface DB table and are at that moment transferable to the ERP.

Based on the EIS-DBI interface the process will end with the provision of the timetickets in the interface DB tables.

3.3.1.8 Expected result

Timetickets are provided in the outbound interface DB table and the ERP collected all open data segments.

3.3.2 Function FRAM-004-002: Detailed planning data for ERP

Background information:

The background information serves to better understand the requirements of FRAM

This information is not relevant for the implementation of the requirements.

The scheduling of operations is done completely in HYDRA. To have basic information at ERP it's possible to provide a defined dataset of information from HYDRA graphic planning (HLS). This information are created based on 3 possible events:

1. Planning of an operation
2. Reallocation of an already planned operation
3. Deallocation of an already planned operation

STD FRAM-004-002-001: EIS-EFD

HYDRA standard interface EIS-EFD will be used.

Data will be transferred via DB interface EIS-DBI instead of standard file based interface.

LIC FRAM-004-002-002: necessary licenses

EIS-EFD (Enhancement for detailed scheduling data to ERP)

3.3.2.1 Activation

Precondition: The functionality is licensed

Each of the possible events is activated based on a configuration.

CONFIG FRAM-004-002-003: Operation planned

Create the HYDRA logging configuration "HLS/EINPLANEN" – see documentation EIS-EFD_81.pdf

CONFIG FRAM-004-002-004: Operation reallocated

Create the HYDRA logging configuration "HLS/UMPLANEN" – see documentation EIS-EFD_81.pdf

CONFIG FRAM-004-002-005: Operation deallocated

Create the HYDRA logging configuration "HLS/AUSPLANEN" – see documentation EIS-EFD_81.pdf

3.3.2.2 Transfer media

Data will be transferred via DB interface, see chapter 3.1.

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3.3.2.3 Data structure

The general data structure of the interface is as follows.

Message type: HY72ADRCK_SC

IDOC type: HY72ADRCK_SC

The segment with data fields is explained in documentation EIS-EFD_81.pdf and the additional EXCEL file for the data mapping. The EXCEL file includes also example data, reference to ERP WinLine datafields and further explanations.

3.3.2.4 Involved objects

Origin of data

- Planning data from HYDRA graphic planning

3.3.2.5 Properties

Data set may be created several times per operation during it's "lifetime" based on different planning steps at HYDRA HLS.

3.3.2.6 Archiving

- Interface data of MLE outbound transactions ("HYSAP-OUTBOUND") are archived after two days and deleted after seven days (configured in HYDRA distribution model).
- If required (especially during the test phase), you can change this interval for every message type.

3.3.2.7 Mode of operation

The mentioned data set is created after the planning in HYDRA HLS is saves. Each changed operation may result in one data set at the interface. The data set structure is always the same, the data itself may change.

3.3.2.8 Expected result

HYDRA HLS planning data is provided from HYDRA to ERP.

4 Appendix

4.1 HYDRA products / licenses

HYDRA licenses:

- Licenses of the following table are required additionally.

HYDRA product	Description	License	Number
Database interface	Additional interface license for the HYDRA database interface.	EIS-DBI	1
ERP interface	Upload of detailed planning data for ERP	EIS-EFD	1
Application Service	Additional license to process order sequences	BDE-APF	1

The additional required licenses will be included in the expanded license overview by Mr. Richarz.